Georgi Dikov

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Experience

Foundation Future Industries Munich, Germany
AI Researcher, Computer Vision *Apr 2025 – Present*

- Leveraged **vision foundation models** (DINO, V-JEPA) and **VLMs** to build **world action models** for advanced robotic **perception and object manipulation** learned from human and robot demonstrations.
- Developed full-stack, real-time **6DoF object pose estimation** pipelines to drive autonomous **pick-and-place** tasks in industrial environments, enabling a **humanoid robot** to operate 3 factory shifts daily with minimal human supervision.
- Engineered a multi-camera rig **calibration** pipeline and a real-time **SLAM** system from scratch, integrated into ROS2 with the rest of the robot perception and control stack.

Qualcomm XR Research Amsterdam, Netherlands
Senior Research Engineer, Computer Vision *May 2021 – Mar 2025*

- Developed core 3D perception algorithms for downstream scene light estimation and geometry reconstruction, encompassing classical and neural **TSDF fusion**, **NeRFs**, and **Gaussian Splatting**, and collaborated with software engineers to integrate these models into production apps.
- Engineered models for **self-supervised monocular depth estimation** and 2D/3D **panoptic segmentation**, quantizing and exporting them for real-time execution on XR edge hardware.
- Mentored PhD interns and drove fundamental research (e.g., on uncertainty-aware depth), yielding more than **10 patents** and **5 top-tier publications** (CVPR, ICCV, ECCV) [1–5].

TomTom Amsterdam, Netherlands
Software Engineer, Machine Learning *Feb 2019 – Apr 2021*

- Developed **semantic segmentation** models for auto-annotation of BEV LiDAR-based HD maps, improving the precision of production models by **4.1%** and thus lowering QC-related costs.
- Supervised an M.Sc. thesis student, resulting in a **patent** and a publication at **ICCV** 2021 [6].
- Built and maintained scalable **MLOps infrastructure** (AWS, Docker, Jenkins) for automated end-to-end HD map production.

Volkswagen ML Research Lab Munich, Germany
Research Intern, M.Sc. Thesis *Dec 2017 – Aug 2018*

- Developed a novel **Bayesian** neural network architecture learning method, boosting the accuracy on small datasets in a few-shot learning setup. Published at **AISTATS** 2019 [7].

Education

Technical University of Munich (TUM) Munich, Germany
M.Sc. Computer Science (GPA: 1.3) *Apr 2016 – Sep 2018*

- **Thesis:** Bayesian Neural Network Architecture Learning, published [7].
- **Exchange:** École Polytechnique (Paris, France). Research work on ML and differential privacy, published [8].
- **Research:** Protein function prediction with 3D convolutional networks, published [9].
- **Coursework:** Machine Learning, Computer Vision, Optimization, and Statistics.

Technical University of Munich (TUM) Munich, Germany
B.Sc. Computer Science *Sep 2012 – Mar 2016*

- **Thesis:** Stereo Vision with Spiking Neural Networks, published [10].
- **Exchange:** Semester at Université Pierre et Marie Curie, Paris, France.

Skills

AI & CV: Object Pose Estimation, Panoptic Segmentation, Depth Estimation, SLAM, 3D Reconstruction, NeRFs, VLAs, WAMs
Languages: Python (Expert), C++ (Familiar), Bash
Frameworks: PyTorch, TensorFlow, OpenCV, Open3D, NumPy, ROS2
DevOps & Infra: Linux, Git, AWS, Docker, CI/CD, Slurm, TensorBoard
Spoken Languages: Bulgarian (Native), English (Fluent), German (Fluent), French (Conversational)

Publications

[1] A. P. Dal Cin, **G. Dikov**, J. Ju, M. Ghafoorian. *AnyMap: Learning a General Camera Model for Structure-from-Motion with Unknown Distortion in Dynamic Scenes.* **CVPR** '25.

[2] F. Langer, J. Ju, **G. Dikov**, G. Reitmayr, M. Ghafoorian. *FastCAD: Real-Time CAD Retrieval and Alignment from Scans and Videos.* **ECCV** '24.

[3] X. Shi, **G. Dikov**, G. Reitmayr, T. K. Kim, M. Ghafoorian. *3D Distillation: Improving Self-Supervised Monocular Depth Estimation on Reflective Surfaces.* **ICCV** '23.

[4] J. Ju, C. W. Tseng, O. Bailo, **G. Dikov**, M. Ghafoorian. *DG-Recon: Depth-Guided Neural 3D Scene Reconstruction.* **ICCV** '23.

[5] **G. Dikov**, J. Vugt. *Variational Depth Networks: Uncertainty-Aware Monocular Self-Supervised Depth Estimation.* **ECCVW** '22.

[6] E. Kassapis, **G. Dikov**, D. Gupta, C. Nugteren. *Calibrated Adversarial Refinement for Stochastic Sem. Segmentation.* **ICCV** '21.

[7] **G. Dikov**, J. Bayer. *Bayesian Learning of Neural Network Architectures.* **AISTATS** '19.

[8] A. Di Luzio, A. C. Viana, K. Chatzikokolakis, **G. Dikov**, C. Palamidessi, J. Stefa. *Catch me if you can: how geo-indistinguishability affects utility in mobility-based geographic datasets.* **LocalRec@SIGSPATIAL** '19.

[9] V. Golkov, M. J. Skwark, A. Mirchev, **G. Dikov**, A. R. Geanes, J. L. Mendenhall, J. Meiler, D. Cremers. *3D Deep Learning for Biological Function Prediction from Physical Fields.* **3DV** '20.

[10] **G. Dikov**, M. Firouzi, F. Röhrbein, J. Conradt, C. Richter. *Spiking Cooperative Stereo-Matching at 2 ms Latency with Neuromorphic Hardware.* **Biomimetic and Biohybrid Systems** '17.